



Lotus EcoAI

“RESTORING WATERS. RESTORING FUTURES.”

PRESENTED BY:

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Our Team

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Introduction

- **Lotus EcoAI** is an environmental technology startup dedicated to solving water pollution using autonomous robotics, edge-AI, and green chemistry.



A photograph of a cornfield where the plants are partially submerged in water, illustrating agricultural runoff. The water is calm, reflecting the green leaves of the corn plants. In the background, there are more corn plants and some distant trees under a blue sky with scattered white clouds.

Core problem:

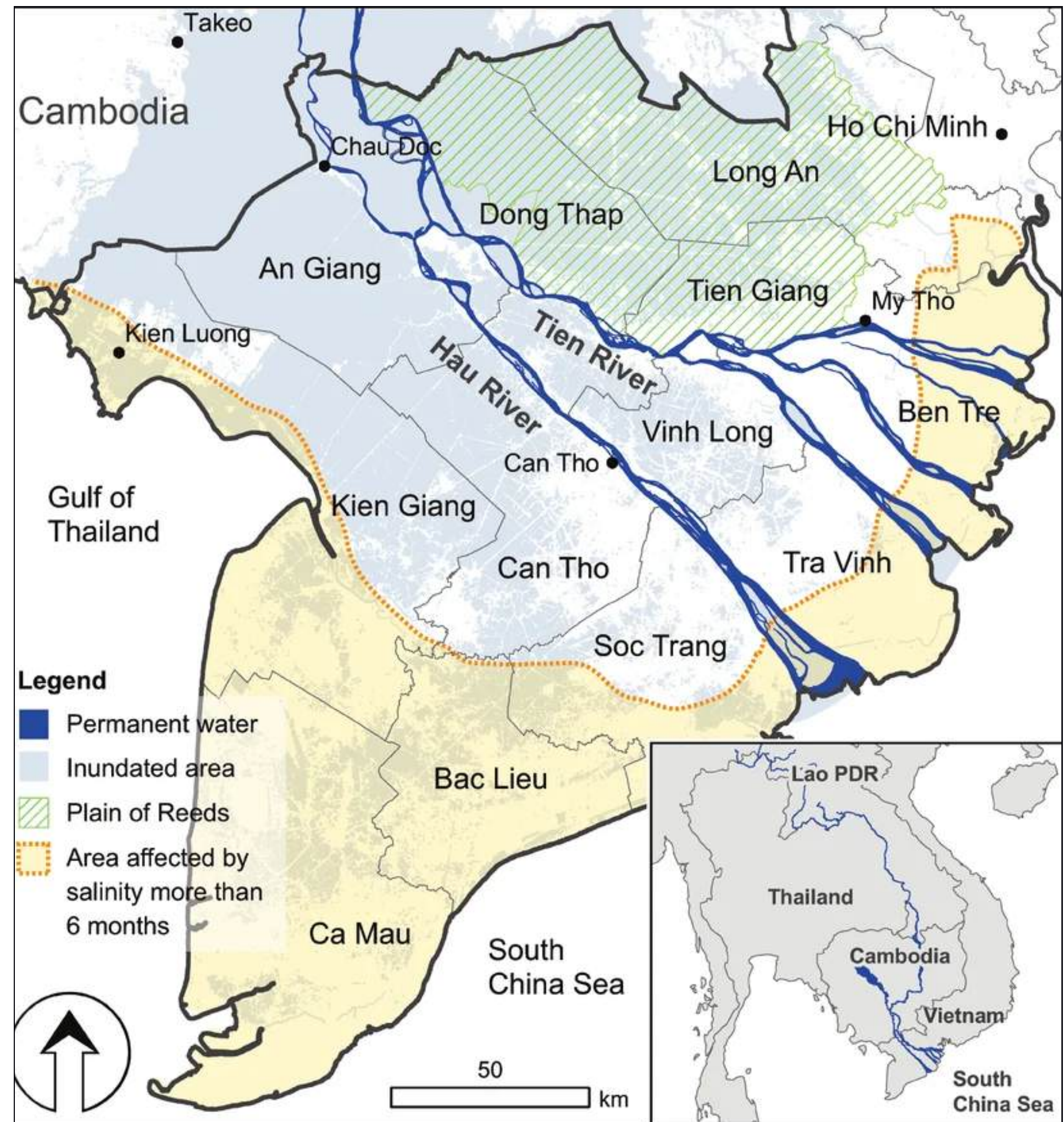
Agriculture runoff

Why this Matters

- The Mekong Delta has over 2.5 million hectares under rice, vegetable and fruit cultivation, using roughly 750,000–1,000,000 tonnes of fertilizer a year UNESCO
- Nitrogen and phosphorus residues from that load flow straight into canals and rivers

75%

of the total phosphorus across the Mekong Basin comes from intensive agriculture



Root Causes

- **Unsustainable Land Management:**
Tillage and deforestation loosen the soil, making it easily washable during rainstorms.

- **Loss of Vegetative Buffer Zones:**
Eliminates physical barriers that would naturally slow down and filter water before it enters river basins.



01. It Starts on the Farm

- Intensive Fertilizer Usage: Farmers over-apply synthetic nitrogen and phosphorus fertilizers, leaving excess chemicals sitting in the soil.



02. It Comes from Everywhere

- Not a single pipe — miles of open fields draining at once
- Nonpoint source pollution: no fixed origin, no measurable outlet
- This is known as a non-point source runoff



What's actually Happening?



03. Nobody Regulates It

- A factory outflow has an address, a permit, and an owner you can fine
- Diffuse runoff has none of those — no single party is responsible → much harder for governments to monitor and enforce than a single factory pipe.



What are the **Consequences**

Severe Eutrophication

- Explosive algal blooms on the water's surface, blocking sunlight from underwater plants.

Toxicity & Bioaccumulation

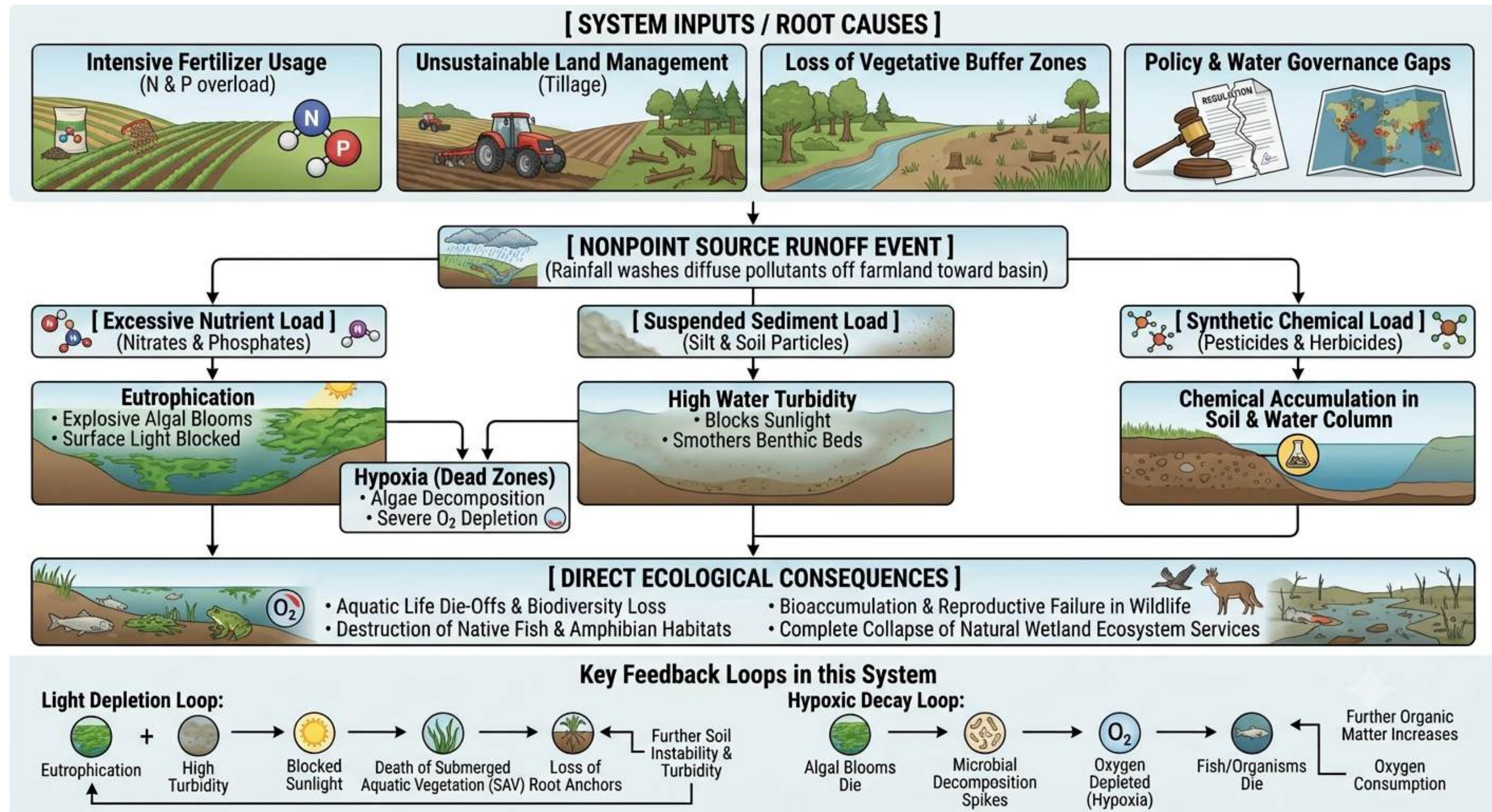
- Pesticides and herbicides build up in water and animal tissues, causing reproductive failure.

Widespread Hypoxia

- Algal blooms die and decompose, bacteria consume all the dissolved oxygen → aquatic "dead zones"

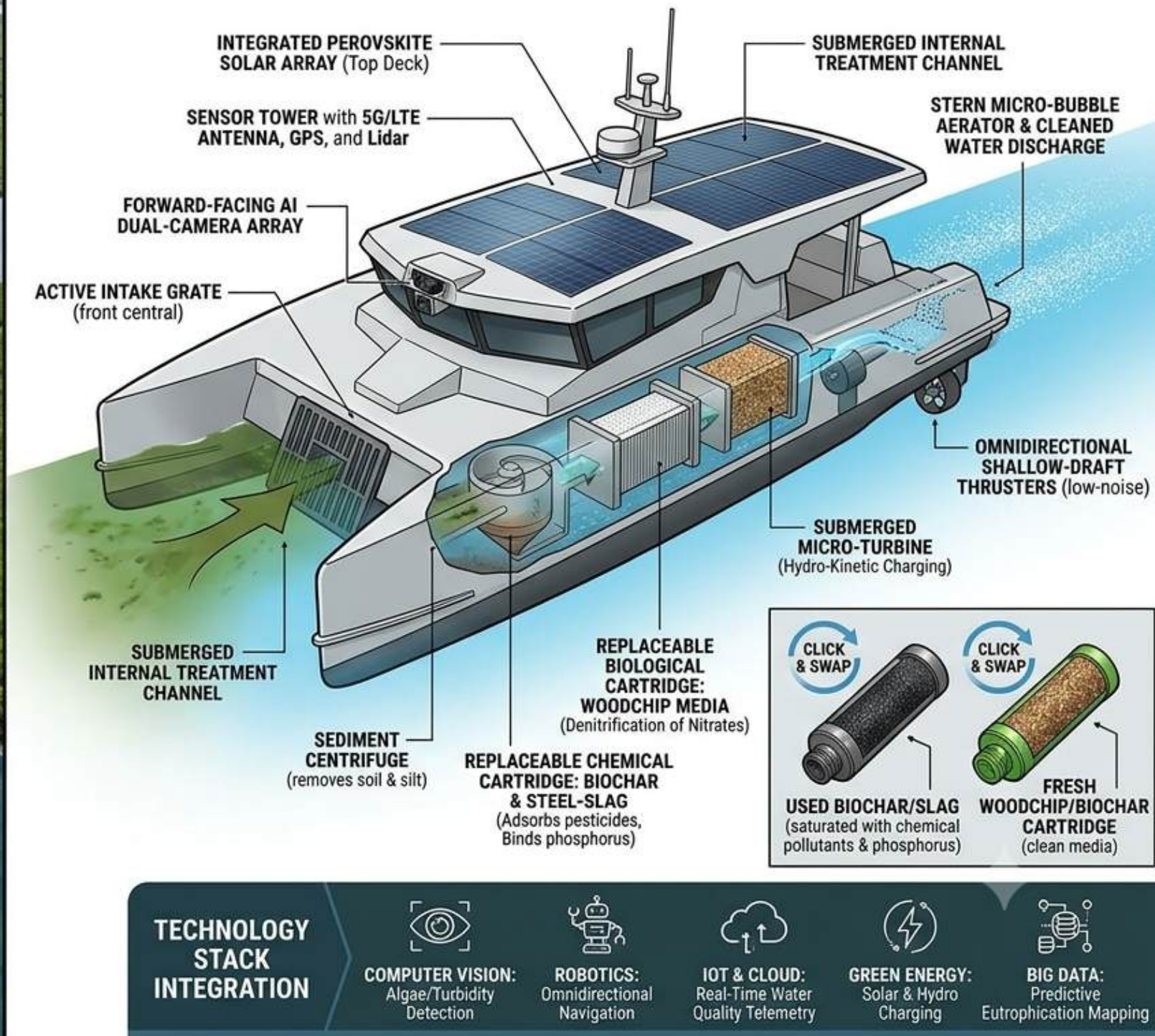
Sedimentation & Turbidity

- Loose silt blankets the wetland floor, clouding the water, smothering fish eggs, and destroying underwater plant habitats.



Excess nutrients → Algae bloom → Sunlight blocked → Algae dies and rots
 → Oxygen used up = Dead zone

Meet Aquathrive - ASV



Autonomous Surface Vessel

A solar-powered catamaran that patrols shallow canals and cleans the water as it goes

How it Works

Stage 1 — Trap the dirt

- **Intake & Sediment Centrifuge:**
- As the vessel moves through water as shallow as 10 cm, an active intake scoop pulls water in. An onboard centrifuge separates heavy soil and silt to clear up cloudy water.



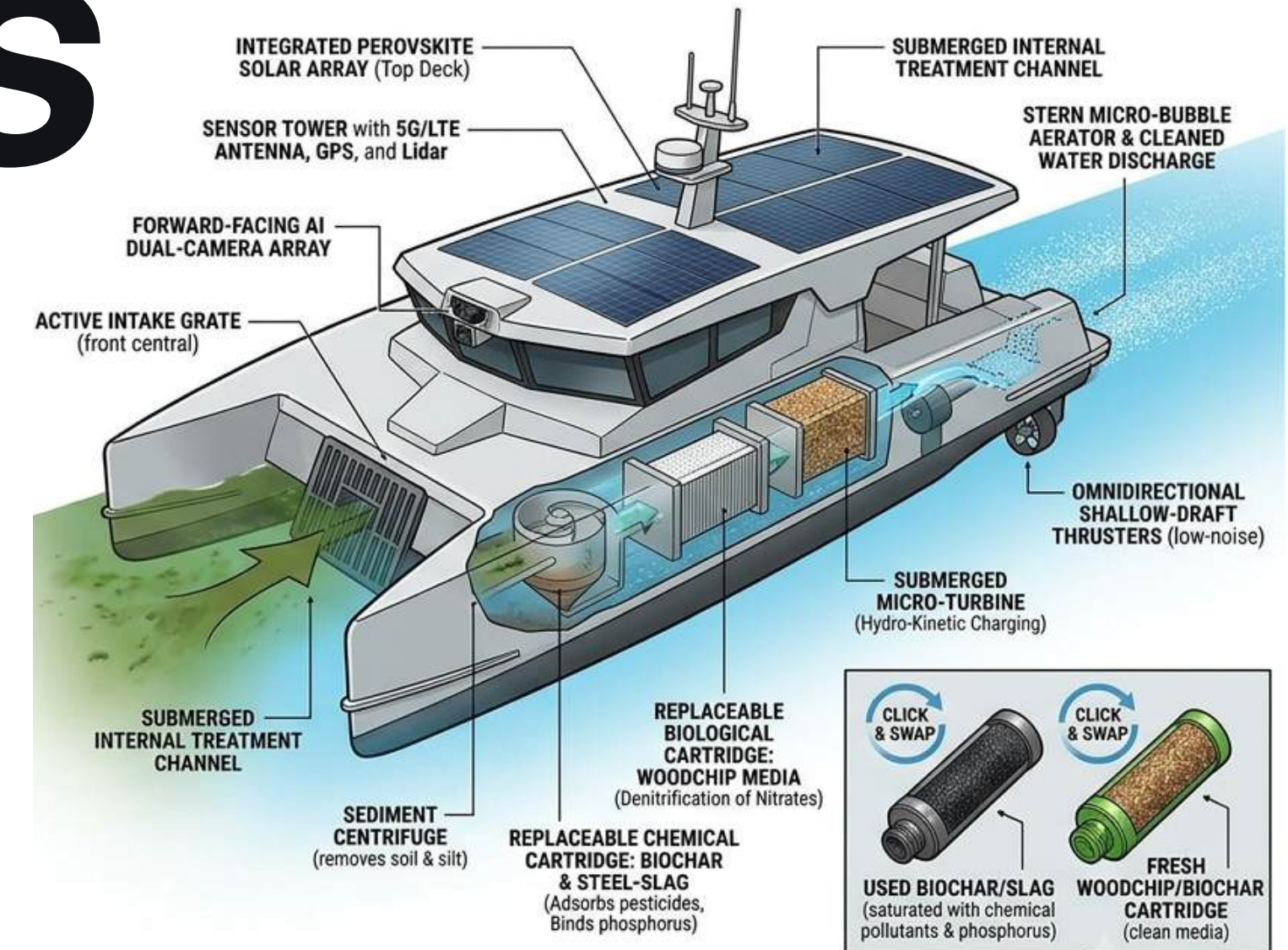
Stage 2 — Filter the chemicals

- **Biological & Chemical Remediation:** Water flows through internal replaceable cartridges:
- **Woodchip Media:** Uses microbial denitrification to convert harmful nitrates into harmless nitrogen gas.
- **Biochar & Steel-Slag:** Steel-slag binds dissolved phosphorus, while high-surface-area biochar adsorbs pesticides and herbicides.



Stage 3 — Put the oxygen back

- **Micro-Bubble Aeration:** Micro-bubble diffusers release fine bubbles from the stern
- Raises dissolved oxygen in stagnant, hypoxic water
- Treats the dead zone, not just its cause



The Technology

AI Vision

- Two cameras spot algae, cloudy water, rubbish and obstacles as the boat moves

Robotics & Actuators

- Two quiet electric motors let it turn in any direction and move through shallow water without stirring up the mud
- Valves adjust how fast water comes in, based on how fast the boat is going

IoT & Cloud

- Integrated 5G/satellite mesh telemetry streaming real-time water metrics — oxygen, pH, temperature, nitrate — to a website.



Green Energy

- Solar panels on top, plus a small underwater turbine that charges the battery when the boat sits in moving water

Predictive Analytics

- Readings from all the boats build a map of where pollution is worst, and warn of algae blooms days before they happen

Business Model

We don't sell boats. We sell clean water. Customers pay a monthly fee per kilometre of waterway managed. We own, deploy and maintain the fleet.

Remediation as a service

- Our main product
- Recurring monthly fee, priced per kilometre managed
- No upfront cost, no maintenance for the customer

Hardware and Cartridges

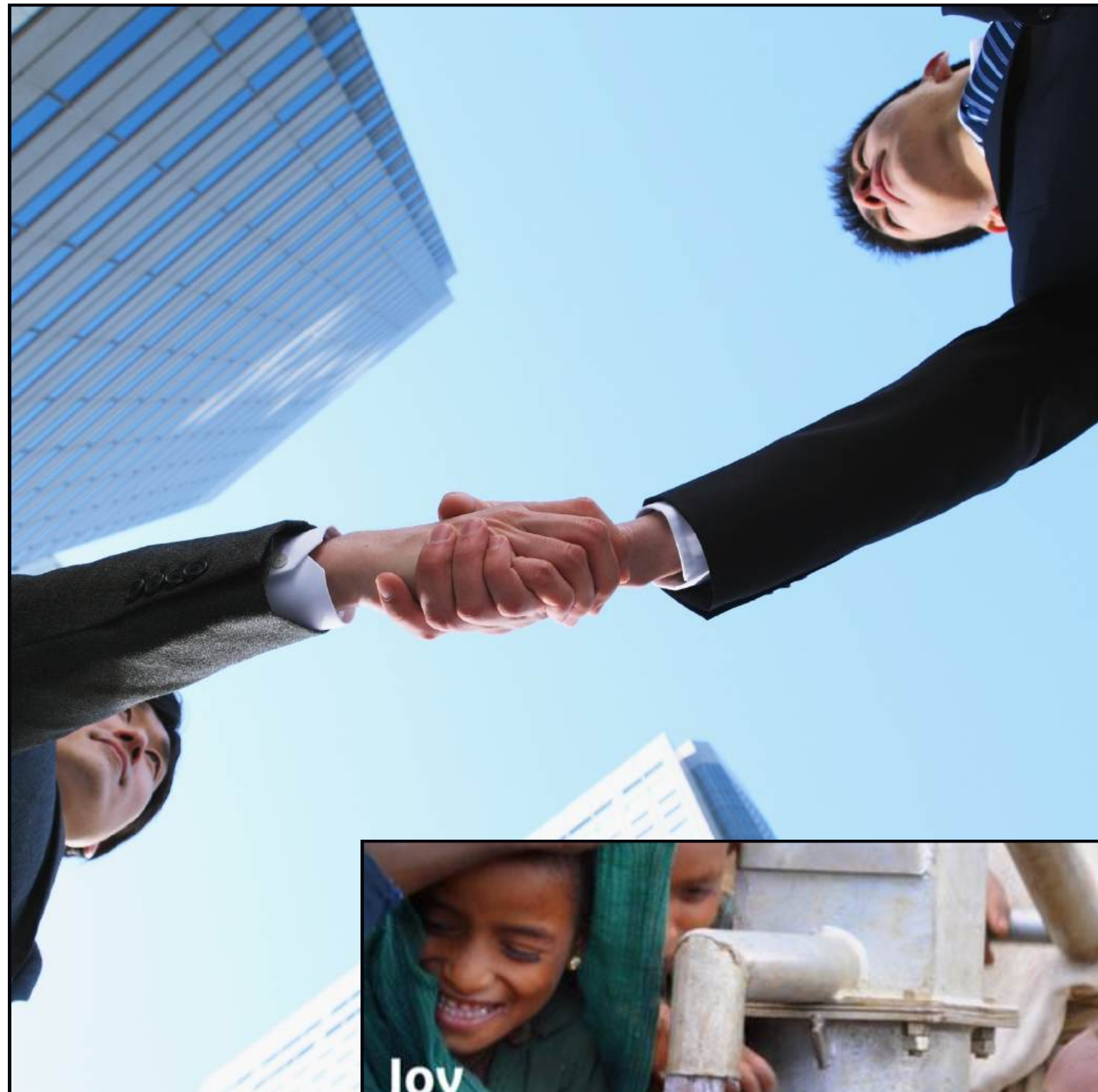
- Buyers who prefer to own can purchase or lease vessels
- Replacement filter cartridges are a repeat purchase

Data Subscription

- Live dashboard showing oxygen, pH, nitrate and turbidity
- Pollution forecasts and the compliance reports companies need for ESG rules

Waste-to-Resource

- Saturated cartridges are collected and processed
- Captured nitrogen and phosphorus become fertilisers
- Sold back to farms



water.org



Target Customers

Farm co-ops and large agriculture

- Required to control runoff — paying us costs less than fines

Water management boards

- Cleaning public canals is already their job and their budget

Industry and mining

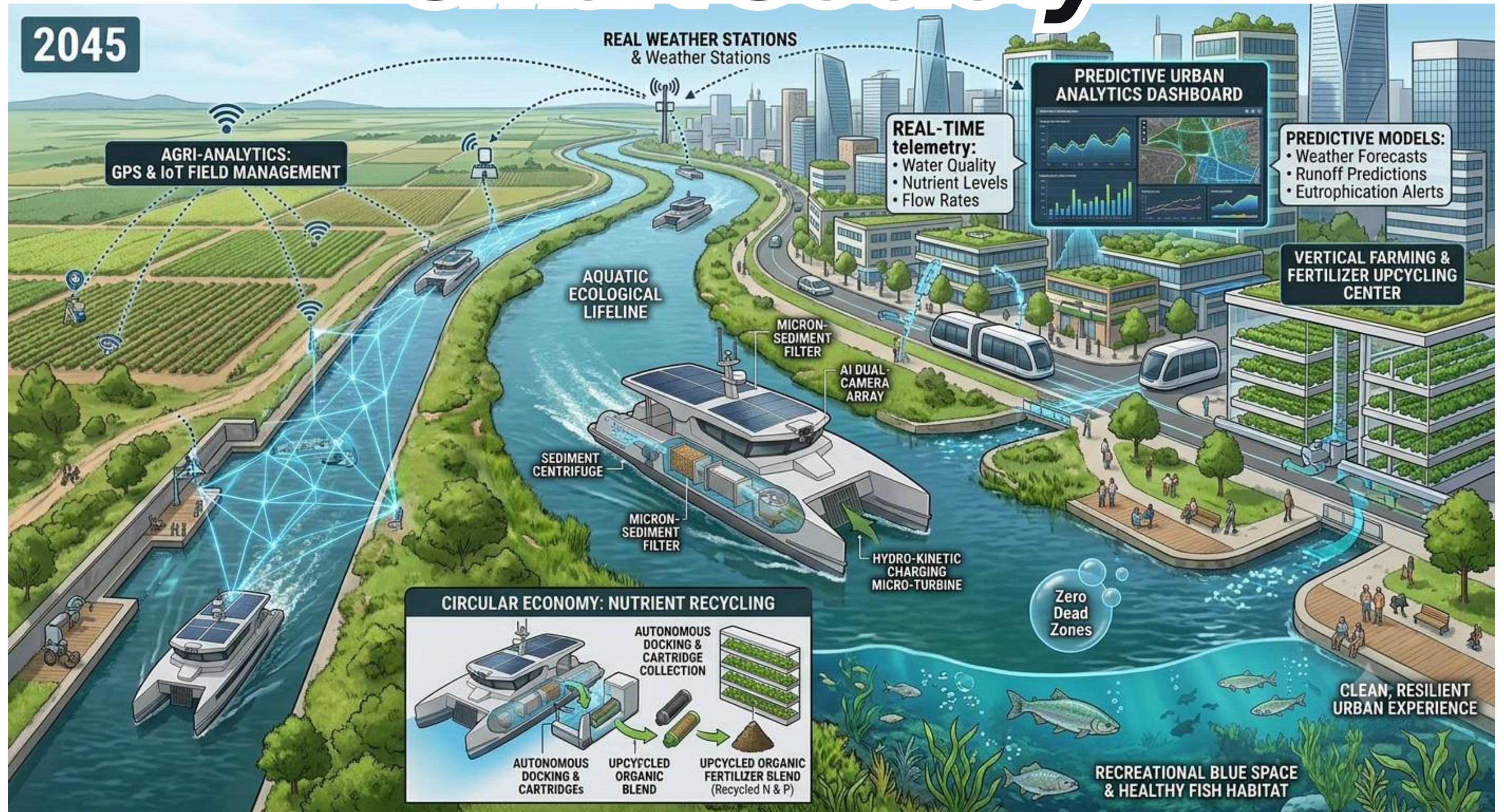
- Waste ponds drain into rivers; they must prove they aren't polluting

NGOs and environmental agencies

- Manage protected wetlands, need evidence their work is doing something



Our Vision of: Smart Society



Reflection:

- **We discovered that: Agricultural runoff is a broken supply chain, not just dirty water.**

Intercepting excess nutrients at the farm edge transforms an ecological liability into a valuable resource before ecosystems collapse.

- **We learned that: Tech succeeds only when aligned with human economic self-interest**

Passive, autonomous systems must handle complexity in the background, turning data into automated compliance and cost savings without adding work for farmers.



Thank You

For your time and
attention

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